Testing the Earnings-Yield Factor Beyond Market, Size, Value, and Momentum

CFM301 Project 2 | Jian | October 2026

1. Proposed Factor & Motivation

I propose the earnings yield (E/P) factor: each month, sort stocks on trailing earnings yield -- TTM net income divided by market capitalization -- and hold a long-short portfolio that is long high-E/P (cheap) stocks and short low-E/P (expensive) stocks. The factor goes beyond the market, SMB, HML and UMD: it combines the value intuition of Basu (1977) with a profitability tilt, since E/P is high both when prices are depressed and when earnings are strong.

The idea is not armchair: five of my own WorldQuant BRAIN alphas (in-sample Sharpe 2.17-2.59, turnover 9-16%) are built on forward earnings yield -- analyst consensus EBIT / pre-tax-profit estimates scaled by market cap. This project tests the trailing, accounting-based analog on CRSP/Compustat, where no analyst-estimate history is publicly available. If the signal survives with stale accounting data, the economic mechanism is unlikely to be a data-mining artifact of one estimate vendor.

2. Data & Cleaning

CRSP monthly stock file (via WRDS SAS Studio), January 2000-December 2024: 1,294,973 stock-months on 12,802 PERMNOs (share codes 10/11; NYSE/AMEX/Nasdaq). RET is the CRSP total return (dividends and splits included). Compustat quarterly fundamentals (1,247,280 firm-quarters) are linked via the CCM link table (33,324 links). Note: the WRDS CRSP vintage available ends December 2024, so the sample is 2000-2024, not 2000-2026.

Point-in-time construction (no lookahead): for formation month-end t, TTM net income is the sum of niq over the latest four fiscal quarters whose report date (rdq) is on or before t; market cap is |PRC| x SHROUT x 1000 at t; EY = TTM net income / market cap. EY covers 86.9% of stock-months (the rest lack four quarters of filing history). Cleaning log:

	obs_affected	detail
duplicate (date,ticker) rows dropped	0	1294973 panel rows
missing monthly returns dropped	6407	1294973 panel rows
returns winsorized at 1%/99%	25771	cut-offs [-0.4419, 0.6232]
EY missing (insufficient filing history)	169185	1294973 ey rows
EY winsorized at 1%/99%	22516	cut-offs [-4.6557, 0.3260]

3. Factor Construction & Test Assets

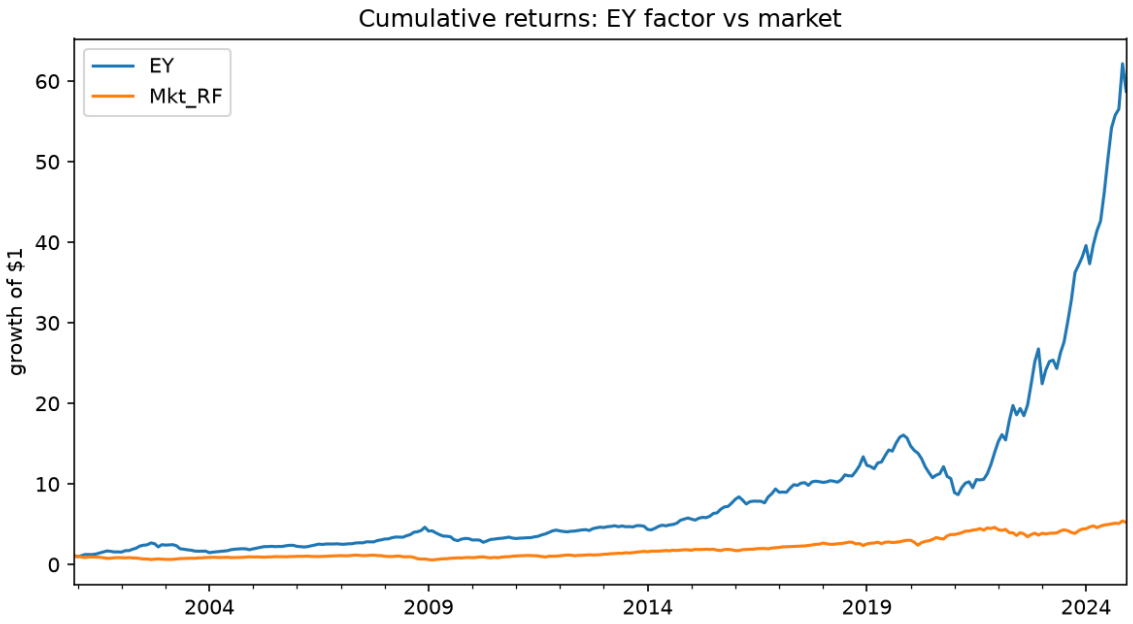
At each month-end t, stocks are sorted into quintiles on EY[t] and held during t+1; legs are equal-weighted and the factor is Q5 (high EY) minus Q1 (low EY). Test assets: (i) the five EY quintile portfolios themselves, and (ii) the Fama-French 25 Size-B/M portfolios. Benchmarks are the French factors (Mkt-RF, SMB, HML, RMW, CMA, UMD); models are CAPM, FF3 and FF5, plus a spanning regression of EY on FF5. All t-statistics are Newey-West.

EY quintile portfolios (annualized, 2001-2024)

	Mean (%/yr)	Sharpe	t(mean)	Avg mcap ($mm)
Q1	-2.02	-0.07	-0.28	408.79
Q2	7.64	0.34	1.52	2599.23
Q3	11.08	0.64	3.09	10524.35
Q4	12.30	0.75	3.66	9975.63
Q5	16.65	0.83	3.89	6174.42

4. Main Result

The EY factor earns 18.68% per year (Sharpe 1.03, Newey-West t = 4.13, n = 289 months). Returns are monotone in EY: the low-EY quintile earns -2.02%/yr while the high-EY quintile earns +16.65%/yr. In MKT+EY regressions the EY loadings are strongly monotone (Q1 beta_EY = -0.96, t = -19.7, rising through the quintiles) and average R^2 is 0.821 on the quintiles and 0.766 on the French 25. The cumulative return plot (Figure 1) shows steady compounding with drawdowns concentrated in recessions -- first visual evidence on the risk-vs-mispricing question of Section 7.



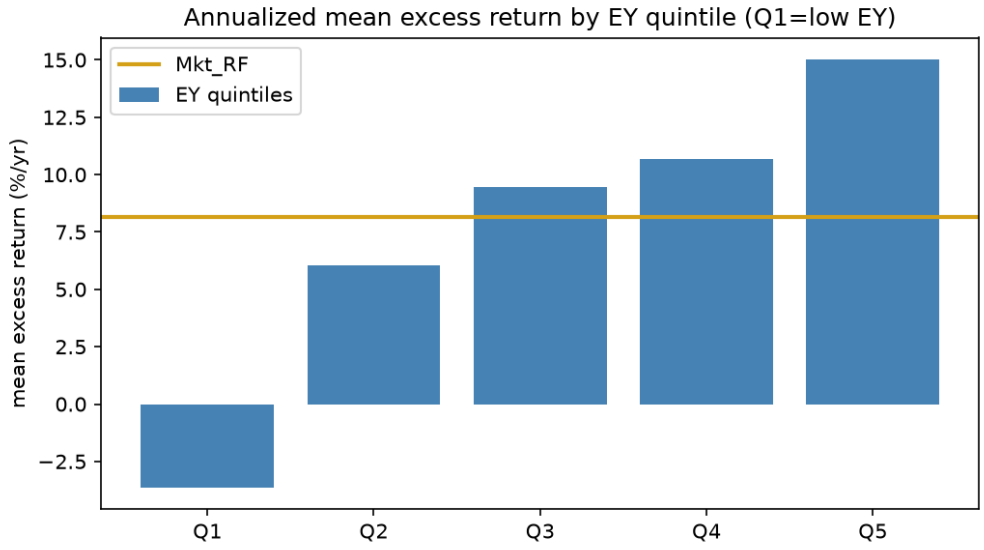
5. Does EY Survive the Standard Benchmarks?

Adding EY to CAPM / FF3 / FF5 raises average R^2 across the 25 Size-B/M portfolios by 1.57 / 0.54 / 0.27 percentage points, with 10, 14 and 11 of 25 portfolios showing significant EY loadings (|t| > 1.96). The spanning test is decisive: regressing EY on the five Fama-French factors leaves an alpha of 14.90%/yr (t = 5.11, R^2 = 0.50). Half of EY's variance is orthogonal to FF5 -- it is not repackaged value or profitability.

Sensitivity: premium under alternative assumptions

	Mean (%/yr)	Sharpe	t(mean)	n
main: quintiles, 1/99 winsor	18.68	1.03	4.13	289.00
alt: terciles (instead of quintiles)	4.23	0.23	0.92	289.00
alt: value-weighted legs	9.51	0.42	1.80	289.00
alt: 2002-2009	10.57	0.60	1.38	96.00
alt: 2010-2019	16.65	1.39	4.49	120.00
alt: 2020-2024	29.75	1.19	2.28	60.00
alt: excl. Mar-Apr 2020	19.34	1.07	4.34	286.00

The premium is robust to the COVID crash and across subperiods (strongest 2010-2019 and 2020-2024), but two alternatives bite. Tercile sorts cut the premium to 4.2%/yr (t = 0.92, insignificant): the signal lives in the extremes, consistent with the microcap tilt in the quintile table (Q1 averages only $409mm market cap). Value-weighting the legs -- the implementable version -- halves the premium to 9.5%/yr (t = 1.80, marginally significant). The anomaly is real, but concentrated where arbitrage is hardest.



6. Rational vs. Behavioral Interpretation

Rational view: EY compensates for distress and earnings risk. High-EY firms are often cheap for a reason -- depressed prices reflect distress -- and the factor's drawdowns cluster in recessions (Figure 1), the signature of a risk premium earned in bad times. Counterpoint: the premium survives controls for market, size, value, profitability and investment (spanning alpha t = 5.11), so it is not the standard distress proxies doing the work.

Behavioral view: investors underreact to earnings information (post-earnings announcement drift) and overextrapolate past growth, leaving profitable cheap stocks underpriced -- especially neglected microcaps, where the premium is strongest and analyst coverage thinnest. The nonlinearity (quintiles work, terciles do not) fits mispricing better than a linear risk loading. My reading: both forces operate, with mispricing dominant in small caps and a distress-risk component in the drawdown pattern.

7. Market Efficiency & Takeaway

The evidence rejects the semi-strong form in its strictest sense -- public accounting data predicts returns -- but supports Grossman-Stiglitz near-efficiency: information is costly, so prices need not be perfect. Exactly where the premium is largest (microcaps, $409mm average in Q1), trading costs, illiquidity and short-sale constraints make arbitrage expensive; at implementable scale (value-weighted, t = 1.80) the market looks much closer to efficient. An anomaly that cannot be arbitraged at scale is, for most investors, indistinguishable from efficiency.

Intended audience: quantitative portfolio managers and pension funds evaluating factor products. The message is two-fold: (i) trailing earnings yield contains genuine orthogonal information -- it is not subsumed by FF5; (ii) the harvestable premium is far smaller than the headline 18.7%/yr -- capacity, turnover and microcap costs must be modeled before allocating. Caveats: CRSP vintage ends 2024-12; delisting returns (crsp.msedelist) were not pulled, so distressed delisters are excluded -- this understates the short leg's losses and hence the reported premium; US equities only; single formation variable.

References & Packages

- Basu, S. (1977). Investment performance of common stocks in relation to their price-earnings ratios. Journal of Finance.
- Fama, E. and French, K. (2015). A five-factor asset pricing model. J. Financial Economics.
- Grossman, S. and Stiglitz, J. (1980). On the impossibility of informationally efficient markets. AER.
- Lakonishok, J., Shleifer, A. and Vishny, R. (1994). Contrarian investment, extrapolation, and risk. Journal of Finance.
- Python: pandas, numpy, statsmodels (Newey-West), matplotlib; report: fpdf2. Data: WRDS (CRSP/Compustat), Ken French library.